Assignment on Single Row Functions

1. WAQTD DETAILS OF THE EMPLOYEES WHO ARE HAVING MORE THAN 4 CHARACTRES IN THEIR NAME.

```
SQL> select *
  2  from emp
  3  where length(ename) > 4;

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM     DEPTNO
---------- ---------- --------- ---------- --------- ---------- ---------- ----------
      7369 SMITH      CLERK           7902 17-DEC-80        800                    20
      7499 ALLEN      SALESMAN        7698 20-FEB-81       1600        300         30
      7566 JONES      MANAGER         7839 02-APR-81       2975                    20
      7654 MARTIN     SALESMAN        7698 28-SEP-81       1250       1400         30
      7698 BLAKE      MANAGER         7839 01-MAY-81       2850                    30
      7782 CLARK      MANAGER         7839 09-JUN-81       2450                    10
      7788 SCOTT      ANALYST         7566 19-APR-87       3000                    20
      7844 TURNER     SALESMAN        7698 08-SEP-81       1500          0         30
      7876 ADAMS      CLERK           7788 23-MAY-87       1100                    20
      7900 JAMES      CLERK           7698 03-DEC-81        950                    30
      7934 MILLER     CLERK           7782 23-JAN-82       1300                    10

11 rows selected.
```

2. WAQTD DETAILS OF THE EMPLOYEES, IF EMP SALARY IS MORE THAN 3 DIGIT AND LESS THAN 5 DIGIT.

```
  1  select *
  2  from emp
  3* where length(sal) = 4
SQL> /

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM     DEPTNO
---------- ---------- --------- ---------- --------- ---------- ---------- ----------
      7499 ALLEN      SALESMAN        7698 20-FEB-81       1600        300         30
      7521 WARD       SALESMAN        7698 22-FEB-81       1250        500         30
      7566 JONES      MANAGER         7839 02-APR-81       2975                    20
      7654 MARTIN     SALESMAN        7698 28-SEP-81       1250       1400         30
      7698 BLAKE      MANAGER         7839 01-MAY-81       2850                    30
      7782 CLARK      MANAGER         7839 09-JUN-81       2450                    10
      7788 SCOTT      ANALYST         7566 19-APR-87       3000                    20
      7839 KING       PRESIDENT            17-NOV-81       5000                    10
      7844 TURNER     SALESMAN        7698 08-SEP-81       1500          0         30
      7876 ADAMS      CLERK           7788 23-MAY-87       1100                    20
      7902 FORD       ANALYST         7566 03-DEC-81       3000                    20
      7934 MILLER     CLERK           7782 23-JAN-82       1300                    10

12 rows selected.
```

3. WAQTD JOIN THE GIVEN STRING 'HI' TO ALL EMPLOYEES NAME.

```
  1  select concat(ename ,' HI')
  2* from emp
SQL> /

CONCAT(ENAME,
-------------
SMITH HI
ALLEN HI
WARD HI
JONES HI
MARTIN HI
BLAKE HI
CLARK HI
SCOTT HI
KING HI
TURNER HI
ADAMS HI
JAMES HI
FORD HI
MILLER HI

14 rows selected.
```

4. WAQTD DETAILS OF CUSTOMERS WHO HAVE ODD EMPID NUMBER.

```
  1  select *
  2  from emp
  3* where mod(empno,2) != 0
SQL> /

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM     DEPTNO
---------- ---------- --------- ---------- --------- ---------- ---------- ----------
      7369 SMITH      CLERK           7902 17-DEC-80        800                    20
      7499 ALLEN      SALESMAN        7698 20-FEB-81       1600        300         30
      7521 WARD       SALESMAN        7698 22-FEB-81       1250        500         30
      7839 KING       PRESIDENT            17-NOV-81       5000                    10
```

5. WAQTD NAME AND SALARY OF THE EMPLOYEES, IF THE EMPLOYE SALARY IS IN ODD NUMBER AND THEIR EMPNO IS IN EVEN.

```
  1  select ename,sal
  2  from emp
  3* where mod(sal,2) != 0 and mod(empno,2) = 0
SQL> /

ENAME             SAL
---------- ----------
JONES            2975
```

6. WAQTD NAME, JOB OF EMPLOYEES IN UPPER CASE.

```
  1  select UPPER(ename),UPPER(job)
  2* from emp
SQL> /

UPPER(ENAM UPPER(JOB
---------- ---------
SMITH      CLERK
ALLEN      SALESMAN
WARD       SALESMAN
JONES      MANAGER
MARTIN     SALESMAN
BLAKE      MANAGER
CLARK      MANAGER
SCOTT      ANALYST
KING       PRESIDENT
TURNER     SALESMAN
ADAMS      CLERK
JAMES      CLERK
FORD       ANALYST
MILLER     CLERK

14 rows selected.
```

7. WAQTD DNAME AND LOCATION OF EMPLOYEES IN LOWER CASE.

```
  1  select LOWER(dname),LOWER(loc)
  2* from dept
SQL> /

LOWER(DNAME)   LOWER(LOC)
-------------- -------------
accounting     new york
research       dallas
sales          chicago
operations     boston
```

8. WAQTD NAME OF EMPLOYEE IN LOWER CASE AND JOB SHOULD BE INTIAL LETTER IN UPPER REMAING LETTERS IN LOWER, WHO ARE HAVING COMMISSION

```
  1  select LOWER(ename),INITCAP(job)
  2  from emp
  3* where comm is not null
SQL> /

LOWER(ENAM INITCAP(J
---------- ---------
allen      Salesman
ward       Salesman
martin     Salesman
turner     Salesman
```

9. WAQTD ROUNDOFF SALARY TO THE NEAREST VALUE IN THE EMPLOYEE TABLE.

```
  1  select ROUND(sal)
  2* from emp
SQL> /

ROUND(SAL)
----------
       800
      1600
      1250
      2975
      1250
      2850
      2450
      3000
      5000
      1500
      1100
       950
      3000
      1300

14 rows selected.
```

10. WAQTD ROUNDOFF THE SALARY TO LOWEST VALUE IN EMPLOYEE TABLE.

```
  1  select TRUNC(sal)
  2* from emp
SQL> /

TRUNC(SAL)
----------
       800
      1600
      1250
      2975
      1250
      2850
      2450
      3000
      5000
      1500
      1100
       950
      3000
      1300

14 rows selected.
```

11.WAQTD DETAILS OF THE DEPARTMENT TABLE WHO HAVE MORE THAN 7 CHARACTER IN THEIR LOCATION.

```
SQL> select *
  2  from dept
  3  where LENGTH(loc) > 7;

    DEPTNO DNAME          LOC
---------- -------------- -------------
        10 ACCOUNTING     NEW YORK
```

12.WAQTD NAME OF THE EMPLOYEES WHO HAVE LAST CHAR AS 'H' IN THEIR NAME.

```
  1  select ename
  2  from emp
  3* where SUBSTR(ename,-1) = 'H'
SQL> /

ENAME
----------
SMITH
```

13. WAQTD NAME, SALARY & HIREDATE OF ENPLOYEES, IF EMPLOYEE NAME STARTS WITH CHAR 'J'.

```
  1  select ename,sal,hiredate
  2  from emp
  3* where SUBSTR(ename,1,1) = 'J'
SQL> /

ENAME             SAL HIREDATE
---------- ---------- ---------
JONES            2975 02-APR-81
JAMES             950 03-DEC-81
```

14.WAQTD DETAILS OF DEPT TABLE WHO HAVE 2ND CHAR AS 'A' IN THEIR DEPTNAME.

```
  1  select *
  2  from dept
  3* where SUBSTR(dname,2,1) = 'A'
SQL> /

    DEPTNO DNAME          LOC
---------- -------------- -------------
        30 SALES          CHICAGO
```

15.WAQTD FIRST HALF OF DEPTNAME NAME.

```
  1  select dname,SUBSTR(dname,1,LENGTH(dname)/2)
  2* from dept
SQL> /

DNAME          SUBSTR(DNAME,1,LENGTH(DNAME)/2)
-------------- -------------------------------------------------------
ACCOUNTING     ACCOU
RESEARCH       RESE
SALES          SA
OPERATIONS     OPERA
```

16.WAQTD SECOND HALF OF LOCATION

```
  1  select loc,SUBSTR(loc,LENGTH(loc) / 2 + 1)
  2* from dept
SQL> /

LOC            SUBSTR(LOC,LENGTH(LOC)/2+1)
------------- ----------------------------------------
NEW YORK       YORK
DALLAS         LAS
CHICAGO        CAGO
BOSTON         TON
```

17.WAQTD NAMES AND COMMISION OF EMPLOYEES, IF THEY HAVE CHAR 'I' PRESENT IN

THEIR NAME.

```
  1  select ename,comm
  2  from emp
  3* where INSTR(ename,'I') > 0
SQL> /

ENAME            COMM
---------- ----------
SMITH
MARTIN           1400
KING
MILLER
```

18.WAQTD NAMES, JOB OF THE EMPLOYEES, IF THEY HAVE CHAR 'A' PRESENT ATLEAST TWICE IN THEIR NAME.

```
  1  select ename,job
  2  from emp
  3* where INSTR(ename,'A',INSTR(ename,'A') +1) > 0
SQL> /

ENAME      JOB
---------- ---------
ADAMS      CLERK
```

19.WAQTD ALL THE LOCATION WHOSE DNAME STARTS WITH STRING 'RES'.

```
  1  select loc
  2  from dept
  3* where SUBSTR(dname,1,3) = 'RES'
SQL> /

LOC
-------------
DALLAS
```

20. WAQTD NAME OF THE EMPLOYEES BY REPLACING CHAR 'A' BY 'P' IN THEIR NAME.

```
  1  SELECT REPLACE(ename,'A','P')
  2* FROM emp
SQL> /

REPLACE(EN
----------
SMITH
PLLEN
WPRD
JONES
MPRTIN
BLPKE
CLPRK
SCOTT
KING
TURNER
PDPMS
JPMES
FORD
MILLER

14 rows selected.
```

21.WAQTD DETAILS OF THE EMPLOYEES, IF THEY HAVE CHAR 'A' EXACTLY TWICE IN THEIR JOB.

```
  1  SELECT *
  2  FROM emp
  3* where LENGTH(job) - LENGTH(REPLACE(job,'A','')) = 2
SQL> /

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM     DEPTNO
---------- ---------- --------- ---------- --------- ---------- ---------- ----------
      7499 ALLEN      SALESMAN        7698 20-FEB-81       1600        300         30
      7521 WARD       SALESMAN        7698 22-FEB-81       1250        500         30
      7566 JONES      MANAGER         7839 02-APR-81       2975                    20
      7654 MARTIN     SALESMAN        7698 28-SEP-81       1250       1400         30
      7698 BLAKE      MANAGER         7839 01-MAY-81       2850                    30
      7782 CLARK      MANAGER         7839 09-JUN-81       2450                    10
      7788 SCOTT      ANALYST         7566 19-APR-87       3000                    20
      7844 TURNER     SALESMAN        7698 08-SEP-81       1500          0         30
      7902 FORD       ANALYST         7566 03-DEC-81       3000                    20

9 rows selected.
```

22. WAQTD NAMES, JOB OF THE EMPLOYEES WHOSE NAME STARTS WITH VOWELS. (ALL NAMES ARE IN UPPER CASE)

```
  1  SELECT ename,job
  2  FROM emp
  3* where SUBSTR(ename,1,1) = 'A' or  SUBSTR(ename,1,1) = 'E' or  SUBSTR(ename,1,1) = 'I'or  SUBSTR(ename,1,1) = 'O'or  SUBSTR(ename,1,1) = 'U'
SQL> /

ENAME      JOB
---------- ---------
ALLEN      SALESMAN
ADAMS      CLERK
```

23.WAQTD DETAILS OF THE EMPLOYEES WHOSE JOB ENDS WITH CONSONANTS. (ALL JOB ARE IN UPPER CASE).

```
  1  SELECT *
  2  FROM emp
  3* where SUBSTR(job,-1) not in ('A','E','I','O','U')
SQL> /

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM     DEPTNO
---------- ---------- --------- ---------- --------- ---------- ---------- ----------
      7369 SMITH      CLERK           7902 17-DEC-80        800                    20
      7499 ALLEN      SALESMAN        7698 20-FEB-81       1600        300         30
      7521 WARD       SALESMAN        7698 22-FEB-81       1250        500         30
      7566 JONES      MANAGER         7839 02-APR-81       2975                    20
      7654 MARTIN     SALESMAN        7698 28-SEP-81       1250       1400         30
      7698 BLAKE      MANAGER         7839 01-MAY-81       2850                    30
      7782 CLARK      MANAGER         7839 09-JUN-81       2450                    10
      7788 SCOTT      ANALYST         7566 19-APR-87       3000                    20
      7839 KING       PRESIDENT            17-NOV-81       5000                    10
      7844 TURNER     SALESMAN        7698 08-SEP-81       1500          0         30
      7876 ADAMS      CLERK           7788 23-MAY-87       1100                    20
      7900 JAMES      CLERK           7698 03-DEC-81        950                    30
      7902 FORD       ANALYST         7566 03-DEC-81       3000                    20
      7934 MILLER     CLERK           7782 23-JAN-82       1300                    10

14 rows selected.
```

24. WAQTD FIRST 3 CHARACTERS OF NAME IN LOWER CASE AND REMAINING IN UPPER CASE IN EMPLOYEE TABLE.

```
  1  SELECT LOWER(SUBSTR(ename,1,3))||UPPER(SUBSTR(ename,4))
  2* FROM emp
SQL> /

LOWER(SUBSTR(ENAME,1,3))||UPPER(SUBSTR(E
----------------------------------------
smiTH
allEN
warD
jonES
marTIN
blaKE
claRK
scoTT
kinG
turNER
adaMS
jamES
forD
milLER

14 rows selected.
```

25.WAQTD TO JOIN A STRING 'HAPPY BIRTHDAY' TO EMP NAME, WHO ARE BORN AFTER '01-JAN-1981'.

```
  1  SELECT 'HAPPY BIRTHDAY ' || ename
  2  FROM emp
  3* where hiredate > '01-JAN-1981'
SQL> /

'HAPPYBIRTHDAY'||ENAME
------------------------
HAPPY BIRTHDAY ALLEN
HAPPY BIRTHDAY WARD
HAPPY BIRTHDAY JONES
HAPPY BIRTHDAY MARTIN
HAPPY BIRTHDAY BLAKE
HAPPY BIRTHDAY CLARK
HAPPY BIRTHDAY SCOTT
HAPPY BIRTHDAY KING
HAPPY BIRTHDAY TURNER
HAPPY BIRTHDAY ADAMS
HAPPY BIRTHDAY JAMES
HAPPY BIRTHDAY FORD
HAPPY BIRTHDAY MILLER

13 rows selected.
```

26.WAQTD DETAILS OF EMPLOYEES WHO BORN IN MONTH OF JULY.

```
  1  SELECT *
  2  FROM emp
  3* where TO_CHAR(hiredate,'MONTH') = 'JULY'
SQL> /

no rows selected
```

27.WAQTD NAME AND SALARY ALONG WITH HIREDATE OF THE EMPLOYEES WHO HIRED ON MONDAY...

```
  1  SELECT ename,sal,hiredate
  2  FROM emp
  3* where TO_CHAR(hiredate,'DAY') = 'MONDAY'
SQL> /

no rows selected
```

28. WAQTD DETAILS OF EMPLOYEES WHO HIRED IN THE MONTH OF JANUARY ON

WEDNESDAY AT 11AM.

```
  1  SELECT ename,sal,hiredate
  2  FROM emp
  3* where TO_CHAR(hiredate,'MONTH') = 'JANUARY' AND TO_CHAR(hiredate,'DAY') = 'WEDNESDAY' AND TO_CHAR(hiredate,'HH24:MI;SS') = '11:00:00'
SQL> /

no rows selected
```

29.WAQTD TOTAL EARNING OF ALL THE EMPLOYEES ALONG WITH NAME AND HIREDATE.

(TOTAL EARNING MEANS (SAL+COMM)).

```
  1  SELECT sal + NVL(comm,0),ename,hiredate
  2* FROM emp
SQL> /

SAL+NVL(COMM,0) ENAME      HIREDATE
--------------- ---------- ---------
            800 SMITH      17-DEC-80
           1900 ALLEN      20-FEB-81
           1750 WARD       22-FEB-81
           2975 JONES      02-APR-81
           2650 MARTIN     28-SEP-81
           2850 BLAKE      01-MAY-81
           2450 CLARK      09-JUN-81
           3000 SCOTT      19-APR-87
           5000 KING       17-NOV-81
           1500 TURNER     08-SEP-81
           1100 ADAMS      23-MAY-87
            950 JAMES      03-DEC-81
           3000 FORD       03-DEC-81
           1300 MILLER     23-JAN-82

14 rows selected.
```

30. WAQTD DETAILS OF THE EMPLOYEES WHO HAVE MORE THAN 35YEARS OF EXPERIENCE.

```
  1  SELECT *
  2  FROM emp
  3* where MONTHS_BETWEEN(SYSDATE,HIREDATE)/12 >= 35
SQL> /

     EMPNO ENAME      JOB              MGR HIREDATE         SAL       COMM     DEPTNO
---------- ---------- --------- ---------- --------- ---------- ---------- ----------
      7369 SMITH      CLERK           7902 17-DEC-80        800                    20
      7499 ALLEN      SALESMAN        7698 20-FEB-81       1600        300         30
      7521 WARD       SALESMAN        7698 22-FEB-81       1250        500         30
      7566 JONES      MANAGER         7839 02-APR-81       2975                    20
      7654 MARTIN     SALESMAN        7698 28-SEP-81       1250       1400         30
      7698 BLAKE      MANAGER         7839 01-MAY-81       2850                    30
      7782 CLARK      MANAGER         7839 09-JUN-81       2450                    10
      7788 SCOTT      ANALYST         7566 19-APR-87       3000                    20
      7839 KING       PRESIDENT            17-NOV-81       5000                    10
      7844 TURNER     SALESMAN        7698 08-SEP-81       1500          0         30
      7876 ADAMS      CLERK           7788 23-MAY-87       1100                    20
      7900 JAMES      CLERK           7698 03-DEC-81        950                    30
      7902 FORD       ANALYST         7566 03-DEC-81       3000                    20
      7934 MILLER     CLERK           7782 23-JAN-82       1300                    10

14 rows selected.
```